

Interference

1. White light is incident at an angle 45° on a soap film 4×10^{-5} cm thick. Find the wavelength of light in the visible spectrum which will be absent in the reflected light. Given $\mu=1.2$ for soap film

(Ans: $\lambda=7755 \text{ \AA}$)

2. A light of wavelength $5800 \text{ \AA}$ is incident on a thin film of glass of R.I. 1.5 such that the angle of incidence in the plate is 60° . Calculate the smallest thickness of the plate which will make it dark by reflection.

(Ans: $t_{\min}=2368 \text{ \AA}$)

3. Light travelling in air is incident on a thin parallel film at an angle 57° . If the reflected and refracted rays make an angle of 90° with each other find the refractive index of the film.

(Ans: $\mu=1.539$)

4. A parallel beam of sodium light ($=5890 \text{ \AA}$) strikes a film of oil floating on water. When viewed at an angle of 30° from the normal the 8th dark band is seen. If the refractive index of oil is 1.46, find the thickness of the film.

(Ans: $t=1.72 \text{ \AA}$)

5. A drop of liquid of volume 0.2 cc spreads over the whole surface of a tank of water of area 1 sq. m. forming a thin film. When white light is incident normally on the film a dark band corresponding to the wavelength $5500 \text{ \AA}$ is seen in the spectrum. Find the refractive index of the liquid.

(Ans: $\mu=1.375$)

6. Find the thickness of the soap film which appears yellow ($\lambda = 5896 \text{ \AA}$) in reflection when it is illuminated by white light at an angle of 45° . Given refractive index of the film $= 1.33$.

(Ans: $t = 1308 \text{ \AA}$)

7. A parallel beam of light of wavelength $5870 \text{ \AA}$ is incident on a thin glass plate of refractive index 1.5 such that the angle of reflection into the plate is 60° . Calculate the smallest thickness of the glass plate which will appear dark by reflection.

(Ans: $t = 2396 \text{ \AA}$)

8. White light is incident on a film at an angle of 53.13° and the reflected light shows two consecutive dark bands for the wavelengths $6000 \text{ \AA}$ and $6100 \text{ \AA}$. If the thickness of the film is 0.0017 cm calculate its refractive index.

(Ans: $\mu = 1.33$)

9. Light of wavelength $6000 \text{ \AA}$ falls normally on a thin wedge shaped film of refractive index 1.4 , forming fringes that are 2 mm apart. Find the angle of the wedge.

(Ans: $\theta = 1.07 \times 10^{-4} \text{ radian}$)

10. Interference fringes are produced by monochromatic light falling normally on a wedge shaped film of cellophane of refractive index 1.4 . If angle of wedge is 20 seconds of an arc and the distance between successive fringes is 0.25 cm , calculate the wavelength of light.

(Ans: $\lambda = 61419 \text{ \AA}$)

11. A glass wedge of angle 0.01 radian is illuminated by a monochromatic light of wavelength 6000Å falling normally on it. At What distance from the edge of the wedge will the 10th dark fringe be observed by the reflected light?

(Ans $d = 2 \times 10^{-5}$ m)

12. Light of wavelength 5500Å falls normally on a thin wedge shaped film of refractive index 1.4 forming fringes that are 2.5mm apart .Find the angle of wedge in second

(Ans: 16.2secs)

13. Two plane rectangular pieces of glass are in contact at one edge and are separated at the other end 10cm away by a wire to form a wedge shaped film. When the film was illuminated by light of wavelength 6000Å, 10 fringes were observed per cm. Determine the diameter of wire.

(Ans: $d = 0.03$ mm)

14. Newton's rings are observed in the reflected light of wave length 5900 Å. The diameter of 10th dark ring is 0.5 cm. Find the radius of curvature of the lens used.

(Ans = 1.059m)

15. Calculate the thickness of air film at the 10th dark ring in a Newton's rings system, viewed normally by a reflected light of wave length 500 nm. The diameter of the 10th dark ring is 2 mm.

(Ans = 2.5µm)

16. Two slits separated by a distance of 0.2 mm are illuminated by a monochromatic light of wave length 550nm. Calculate the fringe width on a screen at a distance of 1 m from the slits.

(Ans = 2.75mm)

17. In a Newton's ring experiment, the diameter of the 5th ring is 0.30 cm and diameter of the 15th ring is 0.62cm. Find the diameter of the 25th ring

(Ans=0.8239cm)

18. In a Newton's ring experiment the diameter of the 10th dark ring changes from 1.4cm to 1.27cm when a liquid is introduced between the lens and plates. Calculate the refractive index of the liquid

(Ans: $\mu=1.215$)

19. In a Newton's ring experiment the diameter of 5th dark ring is 0.336cm. Find the radius of curvature of the plano-convex lens, if the wavelength of light is 5890Å. Also find out the radius of 15th dark ring.

(Ans=0.291cm)

20. A thin transparent film of MgF₂ ($\mu_f = 1.38$) is to be coated on a glass microscope lens ($\mu_g=1.5$) to increase the transmission of the normally incident light of wavelength 5893 Å. What minimum thickness should be deposited on the lens surface?

(Ans: $t_{\min}=1.607\text{Å}$)

